

PROGRAMME : Diploma Programme in Information Technology (IF) / Computer Technology (CM)
COURSE : PHP Programming (PHP) **COURSE CODE** : 233219

LEARNING AND ASSESSMENT SCHEME:

LEARNING AND ASSESSMENT SCHEME																	
Learning Scheme						Credits	Assessment Scheme										Total Marks
Actual Contact Hrs./Week			SLH	NLH	Paper Duration		Theory				Based on LL and TSL				Based on Self Learning		
											Practical						
CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
3	1	2	--	6	3	3	Max 30	Max 70	Max 100	Min 40	Max 25	Min 10	Max 25@	Min 10	Max --	Min --	150

IKS Content: Total Learning Hours for Term: 00 Hrs

@ Internal assessment, # External assessment

1.0 RATIONALE:

PHP is a general purpose, server-side scripting language run a web server that is designed to make dynamic pages and applications. PHP as a web development option is secure, fast and reliable. In the growing field of Web technology it is essential for every Diploma pass outs to learn PHP Language to help them build interactive web applications. This course is designed to inculcate web based applications development skills in students using server side scripting with PHP.

2.0 INDUSTRY / EMPLOYER EXPECTED OUTCOME:

The aim of this course is to help the student to attain the following industry identified Outcome through various teaching learning experiences:

- Develop web based applications using server side scripting with PHP.

3.0 COURSE OUTCOMES:

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate the following course outcomes:

CO1 - Develop programs using basic constructs and control statements.

CO2 - Develop programs by applying object oriented concepts.

CO3 - Create forms with validation to collect user's input.

CO4 - Demonstrate the functions to deal with file processing.

CO5 - Perform database operations in MySQL using PHP.

4.0 COURSE DETAILS:

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-I PHP Fundamentals	1a. Introduction to PHP 1b. PHP programming basics 1c. Recall operators and control flow statements 1d. Introduction to array and string function. 1e. Concept of Function.	1.1 Introduction to PHP, PHP Evolution, PHP Installation, PHP Vs. Other Scripting Languages, PHP vs. ASP, PHP vs. JAVA, PHP vs. Perl. 1.2 PHP Program Basics: Variables, constants, Data types, Scope of Variable. 1.3 Operators: Types of operators, operator Precedence, General operations, String operations.	12

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		1.4 Program Flow Control: Decision making Control statements- if, if-else, nested if, switch, break and continue statement. 1.5 Loop control structures- while, do-while, for and for each. 1.6 Arrays: Types- one Dimensional Arrays, associative array, Multidimensional Arrays, Initializing Arrays, Handling Array with Loops, PHP Array Function. 1.7 Strings: String Function, Converting to String, Converting from String, Formatting Text Strings. 1.8 Functions and its types: Built In Function, User Defined, passed by value, passed by reference, Variable Lifetime, Recursion	
Unit-II OOP with PHP	2a. Necessity of OOP approach with Basic terminologies 2b. Concept of constructor and destructor. 2c. Inheritance and its types.	2.1 OOP approach with PHP, Classes and Objects, Encapsulation, Inheritance, Polymorphism, Cohesion & Coupling 2.2 Creating Classes and Objects Access to Properties & Methods, Public Access, Private Access, Protected Access. 2.3 Constructor and Destructor 2.4 Inheritance, Overloading and Overriding, Cloning Object.	07
Unit-III Creating and Validating Forms	3a. Need of HTML forms 3b. Creating form using different controls. 3c. Concept of Regular Expression.	3.1 Introduction to HTML forms, Creating a web page using GUI components. 3.2 Form controls: text box, text area, radio button, checkbox, list and buttons. 3.3 GET and POST methods. Handling User Input With \$_GET[], \$_POST[]. 3.4 Web page validation: Regular Expressions: Concept, use. Types of Regular Expressions, Perl Compatible	08

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		Expressions. Posix Regular Expression.	
Unit-IV File Handling in PHP	4a. Demonstrate Different File and Directory operations. 4b. Uploading files from client.	4.1 Files Operations: Open, Close, Read, Write, Navigate, Copy, Delete, Rename. Determining file attributes 4.2 Operations on Directory, Add, Delete, Read Directories 4.3 Uploading Files from Clients, Uploading Files with POST	08
Unit-V ODBC and MYSQL	5a. Introduction and necessity of Database. 5b. Use of ODBC Database. 5c. Use of MySQL. 5d. Manage data in the database using PHP script.	5.1 Introduction to Database, Concept of Database abstraction. 5.2 Introduction to ODBC, Configuration of ODBC on windows, Connecting to the database, Manipulating Queries. 5.3 Introduction to MYSQL, Connecting to the database, Manipulating Queries. 5.4 Structured Query Language, Data Definition statements, Creating MYSQL Database, create table, describe, Alter table, Drop table, Drop Database. 5.5 Data manipulation & Retrieval statements, Insert, Delete, Update and Select.	10
		TOTAL	45

5.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Unit No.	Unit Title	Distribution of Theory Marks			
		R Level	U Level	A and above Levels	Total Marks
I	PHP Fundamentals	6	6	6	18
II	OOP with PHP	4	4	4	12
III	Creating and Validating Forms	4	4	8	16
IV	File Handling in PHP	2	4	4	10
V	ODBC and MYSQL	4	4	6	14
	TOTAL	20	22	28	70

6.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:

The tutorial/practical/assignment/task should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills (**Outcomes in cognitive, psychomotor and affective domain**) so that students are able to acquire the desired programme outcome/course outcome.

Note: Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in Affective domain** as given in the mapping matrix for this course. Faculty should ensure that students also acquire Programme Outcomes/Course Outcomes related to affective domain.

Sr. No.	Unit No.	LLO	Practical Exercises (Outcomes in Psychomotor Domain)	Hours
01	I	1a. Implement basic constructs of PHP.	Write a program to show the use of following operators used in PHP - Arithmetic, Logical, Comparison and Relational.	04
02	I	2a. Implement decision control statements.	Write a program using switch case for following cases. - Factorial, Prime number.	02
03	II	3a. Use different inbuilt PHP functions.	Write a program use PHP built-in functions: Array functions, String Functions.	04
04	II	4a. Implement user defined function.	Write a program using function to find factorial of number using following function types: User defined functions with pass by value.	02
05	II	5a. Implement user defined function with pass by value.	Write a program to find a raise to b using passing by value with no return type.	02
06	II	6a. Implement user defined function with pass by reference.	Write a program to find area & perimeter of Rectangle using passing by reference Function.	04
07	III	7a. Implement different types of inheritance.	Write a program to demonstrate implementation of different types of inheritance.	02
08	IV	8a. Implement different operations on file.	Write a program for following File operations: read, write operation.	04
09	V	9a. Create database connectivity using ODBC.	Write a program for Employee management Using ODBC connectivity with Access.	04
10	V	10a. Create database connectivity with MYSQL.	Write a program for Database connectivity and management using MYSQL.	02
			Total	30

7.0 SELF LEARNING: (Assignment / Activities for Specific Learning / Skill Development / Online Courses / Micro projects)

Not Applicable

8.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course,

1. Guide students in undertaking micro projects.
2. Demonstrate students thoroughly before they start doing the practice.
3. Encourage students to refer different website and YouTube channels to have deeper understanding of the subject.
4. Observe and monitor the performance of students in lab.
5. Use of online platforms like MOOCs, Swayam, Spoken tutorial, NPTEL, Infosys springboard etc.

9.0 ASSESSMENT METHODOLOGY

Formative assessment: TH

Periodic Test: 30marks

Summative assessment: TH

70 marks Final theory paper.

Formative assessment: PR

25 marks Lab performance, Viva

Summative assessment: PR

25 marks Lab performance, Viva

10.0 LEARNING RESOURCES:

A) Books:

Sr. No.	Author	Title of Book	Publication
01	Rasmus Lerdorf, Kevin.T and Peter M.	Programming PHP	O'Reilly, USA, ISBN-978-1-449-39277-2, 2013.
02	Holzner, Steven	The Complete Reference PHP (Third Edition covers PHP)	McGraw Hill, New Delhi, ISBN-9780070223622, 2008.
03	McGrath, Mike	PHP and MySQL	McGraw Hill, New Delhi, ISBN - 13: 97801259029431
04	Dr. Rajendra Kawle	Advance Web Technology	Devraj Publication, ISBN – 978-93-86492-01-2

B) Software/Learning Websites:

Sr.No.	Link
01	https://www.w3schools.com/php/default.asp
02	https://www.guru99.com/what-is-php-first-php-program.html
03	https://www.tutorialspoint.com/php/
04	https://tutoriahtml.com/en/php-tutorial-introduction/
05	https://www.tizag.com/phpT/
06	https://books.goalkicker.com/PHPBook/
07	https://codecourse.com/watch/php-basics

C) Major Equipment/ Instrument with Broad Specifications:

Sr.No.	Equipment	Specification
1	Desktop Computer	Desktop Computer with basic configuration (preferably i3-i5)
2	Operating system	Windows/Linux
3	Web server	Apache, Nginx, or Microsoft IIS
4	PHP interpreter	PHP 5.3.3.or above
5	Database	Any database tool such as MYSQL, MariaDB or any equivalent tool
6	Web development tool	XAMPP (all-in-one setup packages consisting Apache, PHP and MySQL)

11.0 MAPPING MATRIX OF PO's, CO's and PSO's:

Course Outcomes	Programme Outcomes (PO's)							Programme Specific Outcomes (PSO's)		
	1	2	3	4	5	6	7	1	2	3
CO1	L	-	M	H	-	-	L	-	-	-
CO2	L	L	H	H	-	L	L	-	L	-
CO3	L	M	L	M	-	-	L	-	-	-
CO4	L	M	H	H	-	L	L	L	-	-
CO5	-	M	M	H	L	L	L	M	-	L

H: High Relationship, M: Moderate Relationship, L: Low Relationship